Kinds of cloning

Read the text about cloning and then complete the task on the next page.

Explanation of cloning

A clone is an organism which is genetically identical to another organism. Cloning is a form of asexual reproduction which is widespread in nature. For unicellular organisms and many plants, it is a normal process, such as cell division or vegetative propagation. Mammals can produce identical twins when very young embryos split in two.

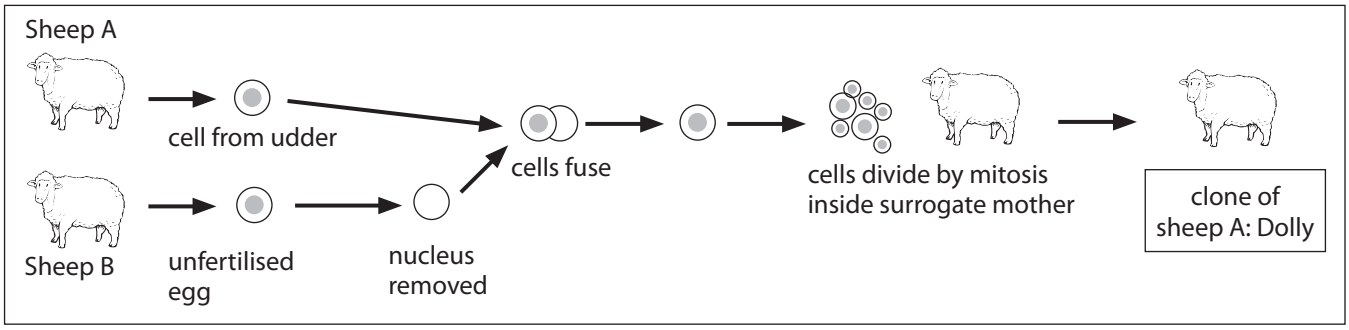
However, artificial cloning is the cause of much debate. There are two types of artificial cloning: embryo splitting and nuclear transfer.

- **Embryo splitting** is similar to the production of twins or multiple births, except it does not happen naturally – it is brought about by a scientist or doctor. In this type of cloning, the clones are the same age.
- During **nuclear transfer**, the complete set of chromosomes of one individual (the genetic parent) is transferred into a stem cell of another individual (the cell donor parent) and carried by a third individual (the host parent). This method allows the chromosomes to be transferred at different times and so the clones can be of completely different ages. For example, a pet cat that has died could be cloned for its owners.

It is also possible to transfer only part of a chromosome of an individual to a complete set of chromosomes of another individual. For example, a strawberry could carry a fish gene to prevent it from freezing at very low temperatures. This method produces **transgenic species**, which could not be produced in nature.

While embryo splitting has a fairly high success rate, nuclear transfer has often resulted in the death of the new clones or in new clones with health or development problems. Dolly, the cloned sheep, was one survivor of 300 clones. The diagram below outlines the process involved in producing Dolly the cloned sheep.

Cloning Dolly



Applications of cloning

In the process of gene farming, transgenic bacteria or animals are used to produce medicines. The advantages of medicines produced by biogenetic manufacturing processes are that they can be obtained in a much purer form than with conventional techniques that involve animal and human intermediate products. Production can be on a large scale and relatively cheap. However, there are also risks to the bacteria and animals due to their transgenic manipulation. People taking those medicines may not respond favourably and could become sick. Careful testing of drugs produced by gene farming is necessary.

In gene targeting or gene knock-out, individuals can be given favourable genes or have unfavourable genes destroyed. This would result in healthier offspring or offspring with certain characteristics. For example, cows could produce milk with a higher protein content. Using this process on animals would also help scientists learn how to cure genetic diseases in people.

During tissue cloning, particular human tissues are grown from the patient's cells rather than using donated tissue or animal tissue. The theory is that the patient's immune system would not reject the cloned tissue because it would not be made from foreign cells. For example, the burnt skin of a patient could be replaced with the patient's own but cloned skin.

Another idea for cloning human tissues involves swapping animal genes with human genes so that the animal grows a particular human organ. This organ would then be transplanted into the human.

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Artificial insemination of farm animals, during which a female animal is fertilised with collected sperm from a particular male, has been occurring for a long time. However, it is a slow process. Usually only one valuable offspring is born at a time. If the chromosomes of a desirable animal could be used to grow clones, many more valuable offspring could be produced. This could result in great commercial and economic advantages. Even dead animals could be cloned.

Concerns about cloning

When cloning occurs in nature, more of already existing organisms are produced. This also happens during embryo splitting and complete nuclear transfer. However, artificial cloning is expensive and wasteful, although it could produce desirable and valuable organisms. As scientists practise cloning, they learn more about genetics, cloning techniques and development. Yet somebody must decide what is desirable and valuable, who should own those organisms, and what happens to the 'old' organisms.

Transgenic species are organisms that do not exist in nature; they are artificially made. It is not known what effect transgenic species could have on nature, if they were released. Once they are released, it may not be possible to capture them again. Potentially, they could interbreed and cause disease and destruction. Many questions about the possible consequences of cloning remain unanswered.

Task

Select one type of cloning and brainstorm possible applications and examples. You could use a mind-map to help organise your ideas. Then write a discussion:

- Start with an introduction in which you describe and explain your type of cloning.
- Develop two elaborated arguments which explain why that type of cloning should occur, and also develop two elaborated arguments against that type of cloning occurring.
- Finally, make a recommendation based on your arguments.

Don't forget to use persuasive language such as *should, must, always, never, only, rarely, extremely, under no circumstances* in order to convince the reader. Use conjunctions such as *on the other hand, however, as well as, but, since, when, because* to link your thoughts together.

To help you structure your discussion, use the scaffold below. Note useful points you could include for each paragraph. These notes will form the basis for your written discussion. Then in your workbook, write your discussion in six paragraphs.

Discussion scaffold

1. Introduction	**2. First argument for**	**3. Second argument for**
4. First argument against	**5. Second argument against**	**6. Conclusion and recommendation**

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